

Inflation Inequality Across the Euro Area

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What We Do

- ▶ Inflation inequality across households increases when relative-price shocks interact with heterogeneous household baskets.
- ▶ We construct monthly indices by income, age, and residence for all 20 euro-area countries, calibrated to official HICP weights.
- ▶ We simulate the effects of temporary VAT cuts, gas and electricity price measures, and fuel rebates on average inflation and inflation inequality.
- ▶ Our approach is close to Jaravel's distributional CPI methodology for the United States.

Source: Jaravel (2026), *Distributional Consumer Price Indices*; author's D-CPI income-quintile series.

Main Messages

- ▶ Inflation differed substantially across households and countries.
- ▶ Housing energy was the main driver of the Q1–Q5 inflation gap.
- ▶ Fiscal policies reduced both average inflation and inflation inequality.

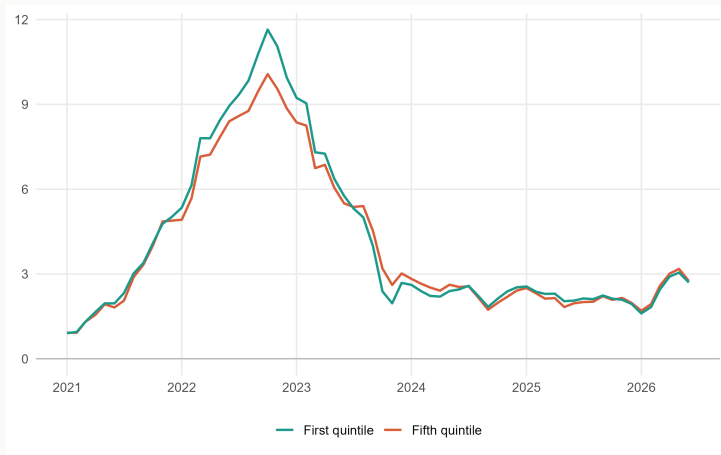
Linking Official Prices to Household Baskets

- ▶ We link monthly Eurostat HICP prices and official item weights to HBS expenditure shares through a harmonized COICOP bridge.
- ▶ HBS tabulations identify baskets by income, age, and residence; microdata add household expenditures, disposable income, demographics, and survey weights.
- ▶ Household expenditure shares come from the HBS; product-level totals are subsequently calibrated to official HICP weights using RAS.

$$P_{g,y,m} = \sum_{j \in \mathcal{J}_y} \hat{\omega}_{j,g,y} \frac{p_{j,y,m}}{p_{j,y,0}}.$$
$$\underbrace{\sum_g \hat{\omega}_{j,g,y}}_{\text{product margin}} = \bar{\omega}_{j,y}^{\text{HICP}}, \quad \underbrace{\sum_{j \in \mathcal{J}_y} \hat{\omega}_{j,g,y}}_{\text{group expenditure-share margin}} = s_{g,y} \quad .$$

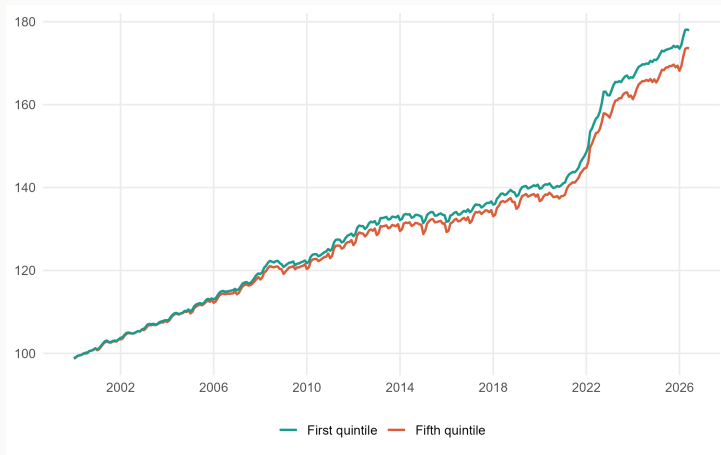
- ▶ Interpretation: baskets are held fixed between HBS waves, so the baseline abstracts from substitution and within-group heterogeneity.

Low-Income Households Faced Higher Inflation During the Energy Shock



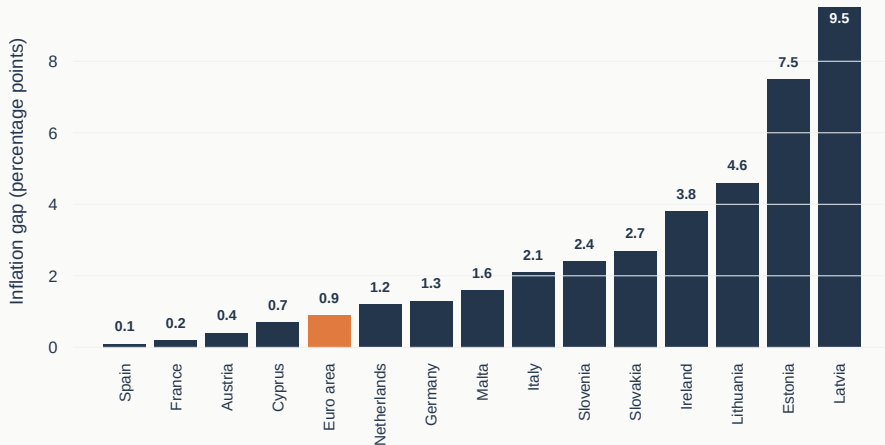
Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Since 2000, Only the 2021–23 Shock Had a Sizable Effect on Inflation Inequality



Source: Eurostat HICP and HBS; authors' calculations. Units: price index, 2000=100.

Heterogeneity in Inflation Gaps Across Countries



Source: Eurostat HICP and HBS; authors' calculations. Units: cumulative Q1–Q5 inflation gap, June 2021–June 2023, in percentage points. Portugal and Finland excluded.

Energy and Food Widened the Gap; Transport Offset It

Product group	Headline	Q1	Q5	Gap
Food and non-alcoholic beverages	4.3	5.2	4.5	0.7
Alcohol, tobacco and narcotics	0.5	0.7	0.4	0.2
Clothing and footwear	0.3	0.3	0.4	-0.1
Housing, rentals and repairs	1.2	0.4	0.6	-0.2
Housing energy	1.5	6.9	5.3	1.5
Transport	1.9	1.6	2.8	-1.2
Other	5.0	0.2	0.3	-0.1
Total	14.7	15.2	14.4	0.9

Source: Eurostat HICP and HBS; authors' calculations. Units: contributions to cumulative inflation, June 2021–June 2023, in percentage points.

Older Households Faced Higher Inflation



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Household-Level Price Indices

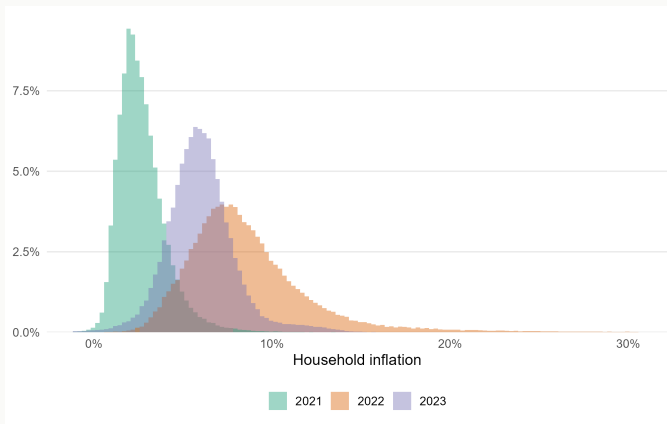
- ▶ We treat each household as an elementary group. Its individual weight $\omega_{j,i,2020}^{\text{HBS}}$ combines its expenditure on item j with the official HICP weight for that item.

$$P_{i,y,m} = \sum_{j \in \mathcal{J}_{i,2020,y}} \omega_{j,i,2020}^{\text{HBS}} \frac{p_{j,y,m}}{p_{j,y,0}}.$$

- ▶ Individual indices measure inflation heterogeneity within income, age, and residence categories, which broad group averages conceal.
- ▶ This within-category dispersion was substantial during the 2021–2023 inflation surge.

Household Groups Hide Household-Level Dispersion

Average inflation and dispersion were both higher in 2022.



Source: Eurostat HICP and HBS microdata; authors' calculations. Units: annual household inflation (x) and household share (y), in percent.

Three Instruments Lowered Retail Energy Prices

- ▶ Euro-area price policies cost **EUR 268 billion**; most were untargeted (**EUR 235.2 billion, or 88%**).
- ▶ **Price caps and tariff shields** limited household electricity and gas prices.
*Exemple : en France, tarifs réglementés du gaz gelés à leur niveau d'**octobre 2021** dès novembre 2021 et hausse du tarif réglementé de l'électricité plafonnée à **4%** en février 2022.*
- ▶ **Temporary VAT and excise-tax reductions** lowered consumer energy prices.
*Exemple : en France, TICFE sur l'électricité ramenée de **22,5 EUR/MWh** à **1 EUR/MWh** de février 2022 à janvier 2023.*
- ▶ **Fuel rebates and regulated discounts** reduced prices at the pump.
*Exemple : en France, remise de **18 centimes par litre** d'avril à août 2022, portée à **30 centimes** de septembre à mi-novembre 2022.*

Source: National policy sources; Sgaravatti et al.; authors' calculations.

Defining the Price-Policy Counterfactual

- ▶ The counterfactual removes these price-based interventions component by component while retaining observed prices for unaffected products.

Example: reversing a temporary VAT cut

$$\tilde{p}_{j,t} = p_{j,t} \frac{1 + \tau_{j,t}^0}{1 + \tau_{j,t}^1}, \quad \tau_{j,t}^1 = \text{policy rate}, \quad \tau_{j,t}^0 = \text{no-policy rate}.$$

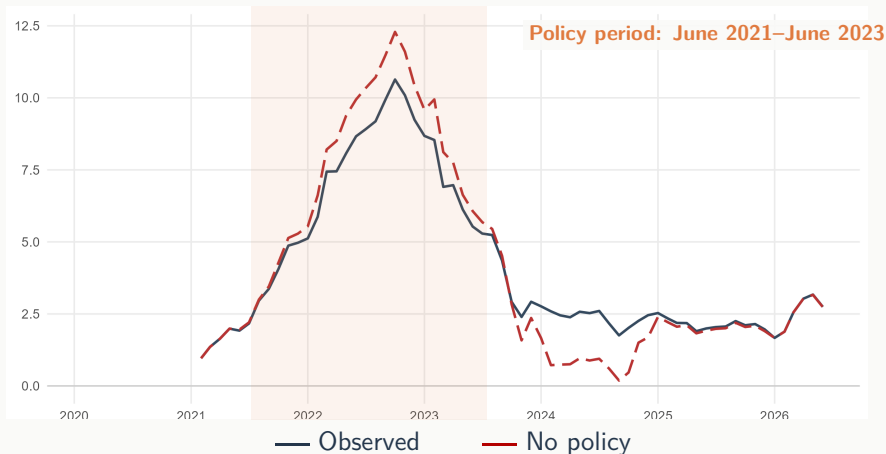
Belgique : une TVA ramenée de 21% à 6% implique

$$\tilde{p}_{j,t} = p_{j,t} \frac{1.21}{1.06} \simeq 1.142 p_{j,t}.$$

j = électricité avec t = mars 2022–décembre 2023.

Source: National policy sources, Sgaravatti et al.; authors' calculations.

Unconventional Fiscal Policies Lowered Inflation by 1.9 pp

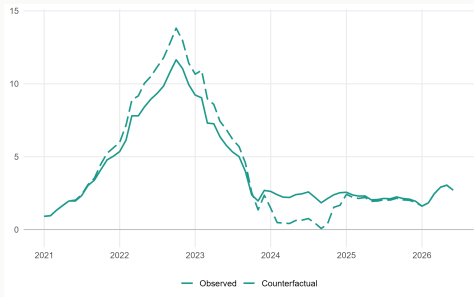


Source: Eurostat and national statistical institutes; authors' calculations. Units: year-on-year inflation, in percent; January 2021–July 2026.

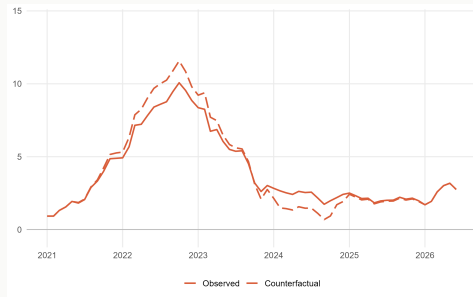
Policies Narrowed the Q1–Q5 Inflation Gap

$$\Delta\text{Gap} = \underbrace{\Delta Q1}_{\text{policy effect}} - \underbrace{\Delta Q5}_{\text{policy effect}} = -1.1 \text{ pp}$$

Q1



Q5



Source: Eurostat and national statistical institutes; authors' calculations. Units: year-on-year inflation, in percent; Jan. 2021–July 2026.

Conclusion

- ▶ The euro-area inflation gap was **0.9 point**, but cross-country differences were large; age and residence also identify vulnerable households.
- ▶ Housing energy and food widened the inflation gap, while transport partly offset it; production networks added indirect exposure.
- ▶ Unconventional fiscal policies lowered cumulative average inflation by **1.9 points** and the Q1–Q5 gap by **1.1 points**.
- ▶ The positive distributional effect came mainly from measures on housing energy, while fuel measures are regressive.

Appendix

Effects of Unconventional Fiscal Policies

Country	Fiscal cost (EUR bn)	No-policy inflation	Avg. effect (pp)	Inflation gap (pp)	Effect of fiscal policies (pp)
Austria	3.8	18.4	-1.2	1.1	-0.7
Estonia	0.2	33.1	-0.1	7.6	-0.1
France	80.5	16.9	-4.7	2.4	-2.2
Germany	60.2	16.5	-0.8	2.2	-0.9
Ireland	1.4	15.0	-0.1	4.0	-0.2
Italy	36.2	16.4	-0.6	2.4	-0.3
Latvia	0.7	28.9	0.0	10.2	-0.7
Netherlands	34.8	19.5	-2.6	2.2	-0.9
Portugal	5.9	17.0	-2.9	-0.8	-1.5
Slovakia	2.3	28.6	-3.3	4.8	-2.1
Spain	23.9	14.3	-2.5	1.3	-1.2
Euro area	268.0	16.6	-1.9	2.0	-1.1

- Negative effects indicate that unconventional fiscal policies reduced inflation or the Q1–Q5 gap.

Source: Eurostat and national statistical institutes; authors' calculations. Units: fiscal cost in EUR billions; cumulative inflation in percent and policy effects in percentage points, June 2021–June 2023.

Rural Households Faced Higher Inflation at the Peak



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Direct Exposure Explains Most of the Energy-Shock Gap

Country	Total	Direct	Indirect	Gap	Gap direct	Gap indirect
Austria	3.3	1.9	1.4	0.7	0.5	0.2
Belgium	4.6	2.6	1.9	1.9	1.7	0.2
Germany	4.1	3.0	1.1	1.5	1.4	0.1
Estonia	4.4	2.1	2.3	2.2	0.8	1.4
Spain	3.1	2.2	0.9	0.2	0.1	0.1
France	3.2	2.3	0.8	0.6	0.6	0.0
Italy	5.1	3.4	1.7	-0.5	-0.4	-0.1
Latvia	4.3	3.1	1.2	3.3	2.3	1.0
Netherlands	3.7	2.9	0.8	1.4	1.4	0.0
Portugal	3.0	2.0	1.1	1.1	0.9	0.2
Slovakia	2.4	1.7	0.7	0.5	0.5	0.0
Euro area	3.9	2.7	1.1	0.8	0.7	0.1

- ▶ A uniform **40%** energy-price shock raises the euro-area Q1–Q5 gap by **0.8 point**; direct exposure accounts for **0.7 point**.

Source: Eurostat HICP, HBS and FIGARO; authors' calculations. Units: simulated cumulative inflation effects, June 2021–June 2023, in percentage points.

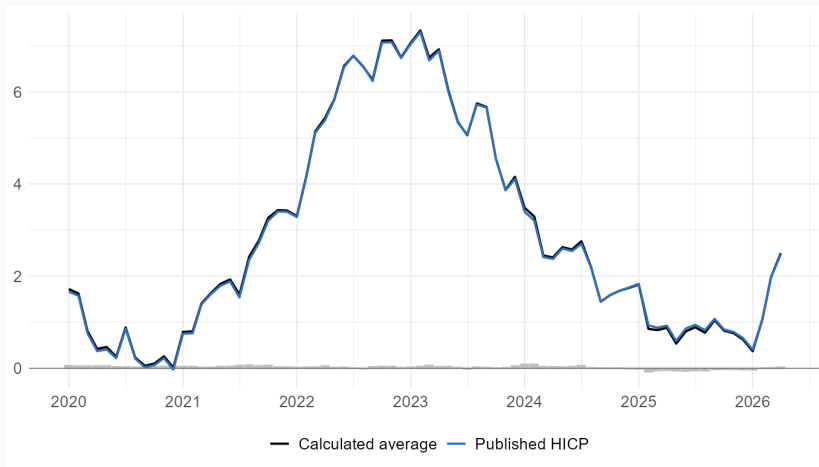
Energy Price Measures Account for the Gap Reduction

Product group	No-policy	Q1	Q5	Gap	Policy effect
Food and non-alcoholic beverages	4.6	5.1	4.4	0.7	0.0
Alcohol, tobacco and narcotics	0.5	0.6	0.4	0.2	0.0
Clothing and footwear	0.3	0.3	0.4	-0.2	0.0
Housing, rentals and repairs	0.5	0.4	0.6	-0.2	0.0
Water, electricity, gas and fuels	7.8	9.7	6.8	2.9	-1.4
Transport	2.5	1.7	3.1	-1.4	0.2
Other	0.3	0.2	0.4	-0.1	0.0
Total	16.6	18.0	16.1	2.0	-1.1

- ▶ Lower gas and electricity prices reduced the gap by **1.4 points**; fuel measures increased it by **0.2 point**.

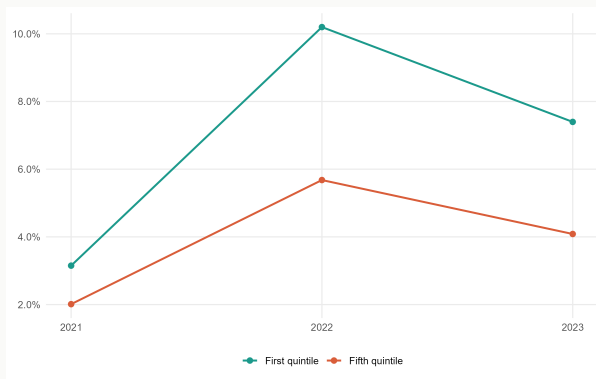
Source: Eurostat and national statistical institutes; authors' calculations. Units: product contributions to cumulative inflation and policy effects, June 2021–June 2023, in percentage points.

RAS-Calibrated Inflation Tracks the Published HICP



Source: INSEE, Eurostat; authors' calculations. Units: year-on-year inflation, in percent.

Inflation Burden Is Larger for Low-Income Households



- ▶ The Q1 burden reached **3.1%, 10.2%, and 7.4%** of net income in 2021–2023, versus **2.0%, 5.7%, and 4.1%** for Q5.

Source: Eurostat HICP and HBS microdata; authors' calculations. Units: annual inflation burden, in percent of household net income.

The Paper Isolates the Household Expenditure Channel

- ▶ **Expenditure:** households buy different baskets, so relative-price shocks generate different inflation rates. This is the channel measured here.
- ▶ **Income:** wages, pensions, and transfers adjust at different speeds, producing unequal changes in real disposable income.
- ▶ **Wealth:** inflation erodes nominal assets and liabilities, transferring wealth from creditors to debtors.
- ▶ Our indices capture heterogeneity in expenditure shares within common COICOP price changes.
- ▶ They do not capture household-specific prices paid, product quality, within-category substitution, or the income and wealth channels.

Source: Main text; presentation framework adapted from the seminar PowerPoint.

Inflation Gaps Differ Sharply Across Countries

Country	Total inflation	Q1 inflation	Q5 inflation	Q1–Q5 gap
Austria	17.2	17.5	17.1	0.4
Belgium	12.3	11.7	12.8	-1.1
Croatia	21.4	21.1	21.7	-0.6
Cyprus	12.0	12.7	12.0	0.7
Estonia	33.0	38.7	31.2	7.5
Finland	12.5	11.5	12.7	-1.2
France	12.2	12.3	12.1	0.2
Germany	15.6	16.5	15.2	1.3
Greece	14.7	15.1	14.5	0.6
Ireland	14.9	17.2	13.3	3.8
Italy	15.8	17.2	15.1	2.1
Latvia	28.9	36.0	26.5	9.5
Lithuania	30.4	33.6	29.0	4.6
Luxembourg	11.4	11.0	11.1	-0.2
Malta	12.7	13.8	12.2	1.6
Netherlands	16.9	17.7	16.5	1.2
Portugal	14.2	13.1	15.4	-2.3
Slovakia	25.4	26.7	24.0	2.7
Slovenia	18.1	19.9	17.5	2.4
Spain	11.8	11.8	11.7	0.1
Euro area	14.7	15.2	14.4	0.9

Source: Eurostat HICP and HBS; authors' calculations. Units: cumulative inflation and Q1–Q5 gap, June 2021–June 2023, in percentage points.

Administered Prices Drive Most of the Gap Reduction

Policy category	Fiscal cost (EUR bn)	Average inflation effect (pp)	Q1–Q5 gap effect (pp)
<i>Instrument classification</i>			
VAT cuts	44.5	-0.5	-0.2
Administered prices	161.0	-1.1	-0.9
Fuel subsidies and rebates	25.4	-0.2	0.1
Other calibrated layer	37.1	-0.2	-0.2
Total	268.0	-1.9	-1.1
<i>Targeted vs. untargeted</i>			
Untargeted price measures	235.2	-1.5	-0.8
Targeted price measures	32.8	-0.4	-0.4
Total	268.0	-1.9	-1.1

- Negative values indicate that policies reduced inflation or the inflation gap.

Source: Bruegel; national sources for VAT food products; authors' calculations. Units: fiscal cost in EUR billions; average-inflation and Q1–Q5-gap effects, June 2021–June 2023, in percentage points.